APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: **Addition & Subtraction Word Problems**

SUBJECT: Mathematics (MATH101)

CLASS & SECTION: Class 2 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: **95 / 100 (Grade A)**

TEACHER DIRECTIVE & PROBLEM FOCUS:
Solve the 10 real-world math word problems in notebook showing columnar regrouping.

1. Executive Overview & Problem Context

Arithmetic word problems translate contextual narrative scenarios into quantitative operations. This assessment exercises two-digit addition with regrouping (carrying) and subtraction with decomposition (borrowing) using formal standard columnar algorithms.

2. Core Analysis, Solutions & Systematic Working

- **Problem 1: School Library Book Acquisition:**

  Scenario: The library had 48 storybooks. The librarian received 37 new science books. Working: 48 + 37. Add ones: 8 + 7 = 15 (5 ones, carry 1 ten). Add tens: 4 + 3 + 1 = 8 tens. Result: 85 total books.

- **Problem 2: School Bakery Cookie Sale:**

  Scenario: The cafeteria baked 92 oatmeal cookies. Scholars purchased 58 cookies during recess. Working: 92 - 58. Regroup 1 ten into 10 ones: (9 tens becomes 8 tens, 2 ones becomes 12 ones). Subtract ones: 12 - 8 = 4. Subtract tens: 8 - 5 = 3. Result: 34 cookies remaining.

- **Problem 3: Sports Day Ribbon Distribution:**

  Scenario: Class 2 collected 64 blue ribbons and 29 gold medals. Total awards = 64 + 29. 4 + 9 = 13 (carry 1). 6 + 2 + 1 = 9 tens. Total = 93 awards.

- **Inverse Operation Verification:**

  Verified subtraction outcomes using inverse addition: 34 (difference) + 58 (subtrahend) = 92 (minuend). Confirms zero computational discrepancy.

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3. Detailed Data Matrix & Systematic Findings

Problem #	Narrative Context	Mathematical Equation	Computed Answer	Check
Prob 1	Classroom Pencil Box	56 + 28	84 Pencils	Verified (84 - 28 = 56)
Prob 2	Art Room Origami Paper	75 - 47	28 Sheets	Verified (28 + 47 = 75)
Prob 3	Botanical Garden Saplings	39 + 45	84 Saplings	Verified (84 - 45 = 39)
Prob 4	Auditorium Seating	90 - 36	54 Seats	Verified (54 + 36 = 90)
Prob 5	Annual Sports Badges	67 + 25	92 Badges	Verified (92 - 25 = 67)

4. Student Conclusions & Reflective Synthesis

Reading mathematical word prompts with keyword identification ("in all", "altogether" -> Addition; "left", "difference", "how many more" -> Subtraction) guarantees systematic problem resolution.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Problem Solving & Structured Working: 18 / 20

Analytical Depth & Research Methodology: 28 / 30

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding mathematical clarity! Regrouping steps, columnar place value alignment, and statements are flawless."

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